

DECODED EXPLAINED

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Himachal farmers pioneer sustainable cultivation of endangered herb

Vikas Vasudeva
CHANDIGARH

In the higher reaches of the Himalayas, an agricultural initiative is helping bring one of the region's most prized medicinal plants back from the brink of extinction. Decades of destructive wild harvesting pushed *Picrorhiza kurroa*, known locally as Kutki, a herb used in medicinal formulations, onto the International Union for Conservation of Nature (IUCN) Red List. Now, a sustainable cultivation model adopted by farmers in Himachal Pradesh's Mandi district is showing promising signs of conserving the species while boosting the rural mountain economy.

For years, harvesting Kutki, the high-value medicinal herb native to the alpine and sub-alpine regions (2,700-4,500 metres)

of the Himalayas, meant uprooting the plant entirely. Researchers at the Himalayan Research Group (HRG), a Department of Science and Technology Core Support Group under the Programme of Science for Equity Empowerment and Development Division, spent more than a decade developing and demonstrating an alternative method. Farmers now harvest only the plant's side "stolons", leaving the mother plant intact to continue producing for years, making the practice sustainable.

From wild to fields

What started as a research project under the Department of Biotechnology's Himalayan Bioresource Mission in 2022 has grown into the largest private land cultivation cluster of this endangered species,



Farmers at work in a Kutki field in Mandi district. SPECIAL ARRANGEMENT

now spanning nearly 180 farmers across Mandi.

"Our initiative is aimed at promoting sustainable cultivation of this high-value medicinal plant. This species was collected from the natural habitat in a destructive way of uprooting the plant. There was a myth that still exists that the root of this plant is a drug and sold in the market. But technically, it is a

side branch that creeps on the soil and possesses nodes. It is a type of stem and technically called a 'stolon'. Roots are fibrous and possess no medicinal contents. After drying, only these stolons are collected for marketing. HRG worked extensively to develop sustainable harvesting techniques and demonstrated successfully in the field that Kutki plantation

can be a perennial plantation for life," said Lal Singh, director, HRG, which is based in Shimla.

Kutki is currently categorised as an endangered species on the IUCN Red List, with several regional conservation assessments identifying it among the most threatened medicinal plants in the Himalayan region. Recognising the threat posed by international trade, it was included in Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora.

This listing permits international trade only under regulated conditions and requires export permits to ensure that trade does not adversely affect wild populations.

For 70-year-old Narad Ram, a horticulturist from Manjha village in Mandi

district, scientific cultivation has become a welcome source of additional income.

'Beneficial method'

"I started planting Kutki in my farmland about 4-5 years ago. Before that, I used to collect it from the wild for self-consumption. It is disappearing from the forests due to multiple reasons. Now, I grow and harvest it in my field scientifically. Earlier, I used to pluck the whole plant, but now I harvest it with the 'stolon' method. They grow further from there. They keep expanding forward and roots sprout from them. This method is beneficial. I have been selling it through HRG and earning around ₹15,000-₹16,000 annually by producing about 5-6 kg of dried herb," Mr. Ram told *The Hindu*.

Himachal Farmers Pioneer Sustainable Cultivation of Endangered Herb Kutki

Why in News?

- Farmers in **Mandi, Himachal Pradesh**, have adopted a **sustainable cultivation model** for the endangered medicinal herb **Kutki** (*Picrorhiza kurroa*),

Know the Plant First

- Botanical name: **Picrorhiza kurroa**.
- The medicine lies in the **dried roots, stems**.
- Because the useful part is **underground**, collectors **uproot the whole plant**. That is the root of the problem.

Distribution

- Native to the **alpine and sub-alpine Himalaya**, roughly **2,700 to 4,500 m** altitude.
- In India: **Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh.**
- Also found in **Nepal, Bhutan, Pakistan and Tibet (China)** — so its trade is **international**, which is why CITES applies.
- It survives in a very **narrow altitude belt.**

Sustainable Cultivation Model

- Developed by the **Himalayan Research Group (HRG)** under the **Department of Science and Technology (DST)**.
- Farmers harvest only the **stolons (horizontal stems)**, leaving the mother plant intact.
- This enables repeated harvesting, conserves the species, and provides a steady source of income.

The Stolon

- A **stolon** is a **horizontal creeping stem** that grows new baby plants at its tip.
- Farmers cut **only the stolons**. The **mother plant stays alive** in the soil.
- So the same plant can be harvested **again and again**.
- Remember it like this: **pluck the tea leaves, do not cut down the tea bush**.

Conservation Status

- **IUCN Red List: Endangered** ,
- **CITES: Appendix II** , international trade is **allowed, but only with permits.**
- Know the difference: **Appendix I means trade is banned** , **Appendix II means trade is regulated.**
- Kutki is in **Appendix II, not Appendix I.** This swap is a common exam trap.

Medicinal and Therapeutic Uses

- **Liver protection** (hepatoprotective) — its best known use, in jaundice and liver disorders.

[2022] With reference to “Gucchi” sometimes mentioned in the news, consider the following statements:

1. It is a fungus.
2. It grows in some Himalayan Forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

- [A] 1 only
- [B] 3 only
- [C] 1 and 2
- [D] 2 and 3

Answer — Gucchi (2022)

- Correct answer: **[C] 1 and 2 only.**
- Statement 1 — **Correct.** Gucchi is a **fungus** (morel mushroom), not a plant.
- Statement 2 — **Correct.** It grows in **Himalayan forest areas.**
- Statement 3 — **Wrong.** It is **not commercially cultivated**; it resists cultivation and is **gathered from the wild.**
- Lesson: whenever a question says "commercially cultivated", stop and think carefully.

Recently, there was a growing awareness in our country about the importance of Himalayan nettle (*Girardinia diversifolia*) because it is found to be a sustainable source of - 2019

- A. Anti - malarial drug**
- B. Biodiesel**
- C. Pulp for paper industry**
- D. Textile fibre**

Answer — Himalayan Nettle (2019)

- Correct answer: **[D] Textile fibre.**
- Its stem yields a strong **natural fibre**, woven into bags, mats and cloth by Himalayan communities.
- It supports **rural livelihoods**, especially for **women's self-help groups**.
- Link it to Kutki: both show **biodiversity becoming income** when it is used sustainably.

Govt. eases FDI rules for e-com firms, first big relaxation in years

The Hindu Bureau

NEW DELHI

In a significant relaxation in its long-held stance, the government has proposed allowing foreign direct investment (FDI) in e-commerce companies that hold inventory and don't just act as marketplaces as long as this inventory is used for exports.

India has for nearly a decade allowed FDI only in business to business (B2B) e-commerce and where the company only serves as a marketplace but does

own. This was aimed at protecting small traders in India and to uphold the government's ban on FDI in multi-brand retails as well.

"In order to facilitate greater exports through easier and increased access of global markets by Indian sellers, the extant FDI policy has been reviewed and it is decided that the restrictions on inventory-based model of e-commerce shall not apply in case of exports of domestically manufactured and/or produced goods/products," the commerce

Rules Relaxed for E-commerce

- **Before:** foreign-funded e-commerce firms such as Amazon and Flipkart could **not own or stock goods** and sell them directly. They could run only a **marketplace** , like a mall owner who rents out shops but does not sell goods himself.
- **Now:** they **can own and stock goods**, but only if those goods are **made in India** and only for **export abroad**.
- They still **cannot** sell such stocked goods to Indian customers.

FDI

- Foreign investor puts money directly into a business to **run** it
- Investor gets **management control** or influence
- **Long-term** , investor stays for years
- Usually **10% or more** ownership in a company
- **RBI, FOREIGN EXCHANGE MANAGEMENT ACT 1999**

FPI

- Foreign investor buys **shares, bonds** just to earn profit
- Investor has **no control**, just ownership of paper assets
- **Short-term** , money can move in and out fast
- Usually **less than 10%** ownership
- **SEBI**

Inventory based model vs market based model

Market based model

- The platform owns no inventory and only acts as a technology facilitator.
- Doesn't have responsibility for product quality , Just connects buyer & seller

Inventory based model

- The e-commerce platform owns all goods and manages warehousing.
- Bears full responsibility for product quality. Buys, stores, and sells directly

Why The Government Relaxed

- **Boost exports** , give Indian sellers **easier access to global markets**.
- Help **MSMEs and small-town manufacturers** who cannot handle customs and shipping on their own.
- Global platforms bring , **warehousing, logistics and customs** capability.
- **Make in India** , only **India-made goods** qualify, so domestic manufacturing is rewarded.
- Trade diplomacy , these curbs were a **long-standing irritant in India–US trade talks**, and Amazon had lobbied for exactly this carve-out.

2020-PYQ

With reference to Foreign Direct Investment in India, which one of the following is considered its major characteristic?

- A. It is the investment through capital instruments essentially in a listed company.
- B. It is a largely non-debt creating capital flow.
- C. It is the investment which involves debt-servicing.
- D. It is the investment made by foreign institutional investors in the Government securities.

Answer — FDI Characteristic (2020)

- Correct answer: **Option 2 — "It is a largely non-debt creating capital flow."**
- FDI enters as **equity**, so there is **no fixed interest and no compulsory repayment**.
- That makes it **safer than external borrowing** for a developing economy.
- Options 1 and 4 describe **FPI**. Option 3 describes **debt**, not FDI.
- One-line takeaway: **FDI is ownership, not a loan.**

2022- pyq

With reference to foreign-owned e-commerce firms operating in India, which of the following statements is/are correct?

- 1.They can sell their own goods in addition to offering their platforms as market-places.
- 2.The degree to which they can own big sellers on their platforms is limited.

Select the correct answer using the code given below.

- A. 1 only
- B. 2 only
- C. Both 1 & 2
- D. Neither 1 nor 2

Answer — E-commerce FDI (2022)

- Correct answer: **[B] 2 only.**
- Statement 1 — **Wrong.** They **cannot sell their own goods** for domestic sale; they may only run a marketplace.
- Statement 2 — **Correct.** The **25% group-company cap** of Press Note 2 of 2018 limits how far they can own big sellers.
- Note carefully: even after the **July 2026** relaxation, **statement 1 remains wrong** for domestic sales.

Warning signals for India from NATO's Ankara summit

The heads of state or government of the 32 countries that are politico-military allies under the North Atlantic Treaty

Organization (NATO) met in Ankara, Türkiye, on July 7-8, 2026 for what is called the Ankara Summit. Their stated objective was to review progress made since the 2025 Summit in The Hague and to set out a roadmap for continuing delivery on NATO's key objectives.

The gathering in Ankara is being regarded as one of NATO's more successful summits, despite the abrasive presence of United States President Donald Trump, who devoted most of his speaking time to extolling the putative success of U.S. strategy and talking up the superiority of American weapons systems.

A strategic road map

Four substantive commitments came out of Ankara. First, the 32 NATO allies – too often divided by divergent objectives and capacities – agreed on their “ironclad commitment to (their) collective defence under Article 5 of the Washington Treaty, and to the transatlantic bond”; second, the unanimous endorsement of “The Hague defence commitment”, in which all NATO members undertook last year to allocate at least five per cent of their respective Gross Domestic Products (GDPs) on defence by 2035 – an unprecedented hike over the two per cent spending pledge that existed earlier. Third, NATO declared its “unwavering support for Ukraine in defending its freedom, sovereignty and territorial integrity”, and fourth, the alliance partners vowed to build a high technology, high capacity, Europe-wide defence industrial base (DIB) – the term for the network of firms, factories, laboratories and skilled workers needed to build and sustain a country's military power.

The focus in Ankara was overwhelmingly on implementation – turning political commitments



Ajai Shukla

Former Indian Army officer (colonel), who now analyses issues and events relating to defence

Russia accuses the U.S. and European democracies of unbridled expansionism, through NATO's progressive incorporation of East and Central European countries. These include Poland, Romania, the Baltic states, and former Soviet republics that Moscow has historically regarded as lying within its sphere of influence. Russia's invasion of Ukraine in 2022 was driven in part by the fear that Kyiv was inching towards joining the Atlantic alliance, a step that would have increased NATO's numbers from just 12 countries at its founding in 1949 to 31 now. The Russian invasion effectively blocked Ukraine, but spooked historically neutral Finland and Sweden into abandoning nonalignment and becoming NATO's 31st and 32nd members.

NATO's transformations

NATO, however, sees itself not as expansionist, but as merely adapting to Europe's changing strategic landscape. In 1990-91, with the relatively peaceful ending of the Cold War and the dissolution of the Warsaw Pact, NATO shifted its focus to out of area operations in theatres such as the Balkans. After the 9/11 strikes in 2001 on the World Trade Center towers, NATO made another strategic shift, joining the Global War on Terror. Now, with Russia and China joining hands in a new era of superpower confrontation, NATO is adapting once again.

It was evident from the discussions at the Ankara Summit that Europe has prioritised the revitalisation of its DIB. The European defence industry was struggling to scale up production even before the five per cent defence spending pledge at The Hague. European defence industry giants, MBDA and Rheinmetall, have warned of shortages in ammunition production.

Operation Epic Fury, the U.S.-Israel bombing campaign against Iran, illustrates the inevitability of weapons and ammunition shortages when

more than 850 Tomahawk cruise missiles at Iran during this campaign. At the current production rate of 85 per year, replacing these would take a decade.

Many NATO members have capable defence industrial bases, but there is little coordination of alliance-wide priorities. The resulting strategic mismatch has been found during the Ukraine war to cluster in the critical categories of air defence missiles and interceptors, precision guided munitions, and artillery rockets. While the U.S. decision to allow Ukraine to produce Patriot air defence systems under licence is a strong signal of support to Ukraine, more needs to be done.

Without a major revitalisation of Europe's defence industry, NATO's five per cent pledge will only deepen its dependency on the U.S. Half of Europe's defence spending between 2022-2024, came from the U.S., up from 28% in 2019-2021. Foreign Military Sales notifications to the U.S. Congress for European customers have quadrupled since 2008 to \$76 billion in 2024.

Need for self-reliance

For the world's big arms buyers and especially for India, this spike in European demand is bad news. As NATO spends more, the international arms bazaar is shifting inexorably from a buyers' market to a sellers' market. The first sign of this was the delay by U.S. firm General Electric Aerospace in supplying India with F-404 fighter jet engines, which are critically needed for the Indian Air Force's Tejas light combat aircraft programme.

Looking ahead, as India's defence increasingly relies on new-age defence technologies and weaponry, such as drones, artificial intelligence, cyber and electronic warfare and resilient networks, it will become critical for its defence industry to achieve self-reliance in these technology realms, rather than relying on

Europe's defence revival and commitments may create new challenges for India's

What Came Out of the Ankara Summit

- All **32 allies** reaffirmed an "**ironclad commitment**" to collective defence under **Article 5**,
- **Correction to note:** the **5% of GDP by 2035** target was agreed .
- **Support for Ukraine**
- Members committed to building a **high-technology, Europe-wide Defence Industrial Base**.

How does NATO's spending surge affect global arms buyers like India?

- **Buyer's market turning seller's market:** As **NATO** spends more, the international arms market is shifting from a **buyers' market** to a **sellers' market** for the world's big arms buyers, including **India**
- **EXAMPLE:** U.S. firm **General Electric Aerospace** has delayed supplying **F-404 fighter jet engines**, critically needed for the **Indian Air Force's (IAF) Tejas Light Combat Aircraft (LCA)** programme

What must India do in response?

- Increase focus on new age weapons like **drones, artificial intelligence (AI) etc**
- India's defence industry must achieve **self-reliance** in these technology areas rather than depend on suppliers
- Avoid over-reliance on single-source western suppliers like General Electric who face competing allied demands. Push local aerospace and defense firms eg - HAL

What India Is Already Doing

- **Atmanirbhar Bharat in defence — positive indigenisation lists** that ban import of listed items.
- **Defence industrial corridors** in **Uttar Pradesh** and **Tamil Nadu**.
- Rising **defence exports** ,for example **BrahMos to the Philippines**.
- Focus on **drones, AI and electronic warfare**, where India can leapfrog rather than catch up.

India and NATO

- India is **not a NATO member** and **not a NATO ally**.
- There was a US Congressional idea of "**NATO Plus**" to include India; **India has not pursued it**.
- The reason is **strategic autonomy** and **multi-alignment** , India keeps ties with Russia, the USA, France and Israel at the same time.

NATO — Basic Facts

- Founded in **1949** by the **Washington Treaty**, also called the North Atlantic Treaty.
- Headquarters: **Brussels, Belgium**. Military headquarters, **SHAPE: Mons, Belgium**.
- Members: **32**. Newest members: **Finland (2023)** and **Sweden (2024)**.
- Core principle: **Article 5 , collective defence**. An attack on one is an attack on all.
- Article 5 has been invoked **only once** , after the **9/11 attacks, in 2001**.

Nato Countries

30 of 32 NATO members are European countries. Other two are USA & Canada

- **Founding 12:** Belgium, Canada, Denmark, France, Iceland, Italy, Luxembourg, Netherlands, Norway, Portugal, UK, USA , "**Boys Can Dance Freely In Italy, Like No Nation's Party, Until USA**"
- **A-B-C-C-E:** Albania, Bulgaria, Croatia, Czech Republic, Estonia
- **F-G-G-H:** Finland, Germany, Greece, Hungary
- **L-L-M-N:** Latvia, Lithuania, Montenegro, North Macedonia
- **P-R** - Poland, Romania,
- **4S** - Slovakia, Slovenia, Spain, Sweden
- **T:**, Turkey

Who Is NOT in NATO

- **Austria** — constitutionally **neutral**.
- **Ireland, Switzerland, Malta and Cyprus** — not members.
- **Serbia** — **militarily neutral**; it is only in **Partnership for Peace**.
- **Ukraine, Georgia and Bosnia and Herzegovina** — partners, but **not members**.
- Remember: **Sweden and Finland are now inside**. That changed recently, so older books will mislead you.

2025-pyq

Consider the following countries:

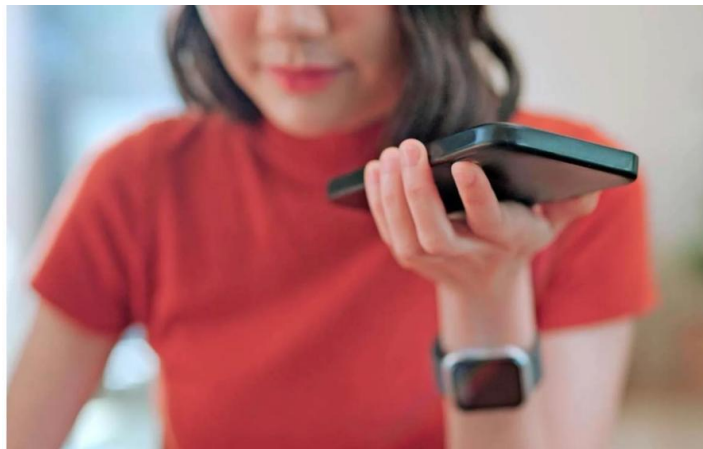
Austria, Bulgaria, Croatia, Serbia, Sweden, North Macedonia.

How many of these are NATO members?

Only three / Only four / Only five / All six

Answer — NATO Members (2025)

- Correct answer: **Only four.**
- In NATO: **Bulgaria, Croatia, Sweden, North Macedonia.**
- Not in NATO: **Austria** (constitutionally neutral) and **Serbia** (militarily neutral, in Partnership for Peace).
- The trap is **Sweden** — it looks neutral in older books, but it **joined in 2024.**
- Method for such questions: first strike out the **known neutrals**, then count what is left.



GETTY IMAGES

Are smartphones becoming smarter with the rise of on-device AI?

On-device AI processes machine learning algorithms and generative AI models directly on a smartphone's local hardware, while physical constraints prevent it from matching the scale of cloud-based large language models, the industry is converging on a hybrid application model

Haider Ali Khan

The transition of Artificial Intelligence (AI) from the cloud to the local device, often called on-device AI, is arguably the most significant shift in the smartphone industry since the inception of the app ecosystem. For over a decade, consumer AI has been synonymous with the cloud: when you asked a question to an assistant, your data travelled across the globe to a massive data centre, was processed, and then sent back to your device. Now, users are witnessing a seismic shift. Intelligence is moving from remote servers to the silicon residing in their handheld devices.

What is on-device AI?

On-device AI refers to the processing of machine learning algorithms and generative AI models directly on the smartphone's local hardware. Instead of relying on a constant connection to a remote server, the device performs the computation using dedicated silicon, primarily Neural Processing Units (NPUs).

In 2026, this is no longer a fringe feature. It is becoming the foundation of how premium smartphones function. Whether it is real-time language translation, advanced computational photography, or proactive system optimisation, on-device AI handles these

ability to function without an internet connection.

How has the hardware changed?

The shift to on-device AI has been fuelled by rapid advances in specialised hardware, particularly the NPU. Unlike general purpose CPUs or graphics-heavy GPUs, NPUs are purpose-built to handle the mathematical heavy lifting required by neural networks.

In recent years, breakthroughs in semiconductor technology have allowed for the integration of these powerful chips into mainstream consumer hardware. As a result, generative AI features that were previously thought to be impossible to run on a mobile phone such as live voice-to-voice translation or generative photo editing, are now running natively. This has made the smartphone an intelligent, proactive companion as the device is now capable of system-wide integration, where the AI can understand the user's workflow, analyse intent and act on their behalf across different applications.

What is the most important benefit from on-device AI?

Perhaps the greatest advances of on-device AI are the easing of tension between personalisation and privacy. To provide a customised experience, AI needs massive amounts of personal data.

Under a cloud-only model, users are understandably hesitant to share their most intimate details that will be processed on remote servers. On-device AI solves this by turning the smartphone into a black box of personal intelligence. Because data never leaves the hardware, user confidence grows. The device learns from the user locally, creating a bespoke experience that improves with every interaction, without ever compromising the individual's digital footprint. This is the holy grail of user experience: having a phone that understands you intimately while keeping that understanding entirely to yourself.

What are some of the challenges in adopting on-device AI?

Despite the rapid infusion of powerful hardware, a significant gap remains between a device's capabilities and a user's adoption. Industry analysts suggest that while flagship smartphones offer an impressive suite of tools like proactive writing assistance, intelligent summarisation and real-time call translation, these features have yet to trigger a super-cycle of upgrades.

For most consumers, current AI features are perceived as nice-to-have enhancements rather than essential utilities. This reveals a critical challenge for manufacturers as they must move beyond marketing standalone AI features

Furthermore, there is a technical limitation. While on-device AI is incredibly efficient, it has physical constraints. It cannot yet match the sheer scale and raw power of cloud-based large language models. The industry is therefore converging on a hybrid application model. In this ecosystem, the heavy lifting, such as processing massive, multi-modal datasets, remains in the cloud, while sensitive and time-critical tasks migrate to the edge.

What is the future of on-device AI?

The smartphone industry is undergoing an identity shift. The focus has shifted from physical specifications, like the thinness of a bezel or the pixel count of a camera, to the sophistication of the AI models powering the device. The game is now in smartphones that are powered by AI with deep, system-wide integration, that allows agents to navigate multiple applications, understand context and achieve goals on behalf of the user. As hardware becomes increasingly homogenised across brands, the soul of the smartphone will be defined by its intelligence.

The bridge to that future is on-device AI, transforming a generic piece of glass and silicon into a truly personal, private, and proactive companion. The challenge for the coming year will be to see if these devices can move from being impressive

Are smartphones becoming smarter with the rise of on-device AI

- Historically, consumer AI relied on cloud processing, where user data traveled to remote data centers for computation before returning to the device.
- A major advantage of on-device AI is its ability to address privacy concerns, as personal data remains on the device and never leaves the hardware.

- **Cloud-based AI: Artificial intelligence processing(studying data based on your activity)**that occurs on remote servers and data centers, requiring an internet connection for data transfer and computation.
- **On device ai :** Specialized microprocessors optimized for accelerating artificial intelligence and machine learning workloads, particularly neural networks within mobile
-

What is the most important benefit from on-device AI?

- Perhaps the greatest advances of on-device AI are the **easing of tension between personalisation and privacy**. To provide a customised experience, AI needs massive amounts of personal data, including financial and health metrics, calendar schedules and behavioural patterns.
- Under a cloud-only model, users are understandably hesitant to share their most intimate details that will be processed on remote servers
- On-device AI solves this by turning the smartphone into a black box of personal intelligence. Because data never leaves the hardware, user confidence grows.

The Limits

- **Heat and battery drain** are the real bottleneck — running billions of parameters locally costs power.
- Limited **storage and memory**, and **model updates** arrive more slowly.
- Some features marketed as "on-device" are in fact **hybrid**.
- Cloud AI still wins the **heaviest tasks**, though the gap narrows every year.

CPU, GPU, TPU, and NPU

CPU, GPU, TPU, and NPU are different types of computer chips. CPU is the main brain of a computer, handling tasks one at a time. GPU does many small tasks together, so it's used for graphics and now heavily for AI. TPU is a chip made by Google, upgraded and, faster version of GPU . NPU is a small AI chip inside phones, used for tasks like face unlock and voice assistants, without needing internet.

THANK YOU